588. $\frac{d}{dx}\left(3+5x^2\right)^{1/4}=$

A. $\frac{1}{4}\left(3+5x^2\right)^{3/4}$

B. $\frac{1}{4}\left(3+5x^2\right)^{3/4}(10x)$

C. $\frac{1}{4}\left(3+5x^2\right)^{-3/4}$

D. $\frac{1}{4}\left(3+5x^2\right)^{-3/4}(10x)$

---

589. If $f'(x)=6x^2-2x+5$ and $f(1)=-3$, then $f(2)=$

A. -19

B. 8

C. 13

D. 22

---

590. The function *f* is given by $f(x)=\frac{\sin x}{x}$. What is the instantaneous rate of change of *f* at $x=\frac{\pi}{2}$?

A. $-\frac{4}{\pi^2}$

B. 0

C. $\frac{2}{\pi}$

D. $\frac{4}{\pi^2}$

---

591. Using a midpoint Riemann sum with subintervals $[1,1.5]$, $[1.5,3]$, and $[3,4]$, what is the approximate distance, in miles, the car traveled from time $t=1$ to time $t=4$?

The table shown gives the speed $s(t)$, in miles per hour, of a car at selected times *t*, in hours.

***t* (hours)**	1	1.25	1.5	2.25	3	3.5	4
$s(t)$ **(miles per hour)**	55	50	65	70	55	60	55

A. 170

B. 175

C. 180

D. 190

---

592. Which of the following statements is true?

Let *f* be the function defined as shown.

$$f(x) = \begin{cases} 1 & \text{if } x < 0 \\ x^2 + 1 & \text{if } 0 \leq x \leq 1 \\ x + 1 & \text{if } x > 1 \end{cases}$$

A. *f* is continuous and differentiable for all *x*.

B. *f* is continuous for all *x* and differentiable except at $x = 1$.

C. *f* is continuous for all *x* and differentiable except at $x = 0$ and $x = 1$.

D. *f* is continuous except at $x = 1$.

---

593. Let *f* be a differentiable function that is decreasing on the interval $(-1, 1)$ and increasing on the intervals $(-2, -1)$ and $(1, 2)$. Which of the following could be the graph of f', the derivative of *f*, on the interval $(-2, 2)$?

A.

- The curve is nearly horizontal as it begins at a maximum at the point with coordinates negative 2 comma 2.
- It then moves steeply downward and to the right. Just as it approaches the x axis, the curve moves more gradually downward and to the right briefly as it crosses the x axis at negative 1.
- From here, it moves steeply downward and to the right again until it levels off and reaches a minimum on the y axis at negative 1.
- The curve then moves steeply upward and to the right. Just as it approaches the x axis, the curve moves more gradually upward and to the right briefly as it crosses the x axis at 1.
- From here, it moves steeply upward and to the right again until it levels off and reaches a maximum at the point with coordinates 2 comma 2.

B.

- The curve begins on the x axis at negative 2 and moves upward and to the right until it levels off and reaches a maximum at the point with approximate coordinates negative 0.9 comma 2.2.
- It then moves downward and to the right, passes through the origin, and levels off as it reaches a minimum at the point with approximate coordinates 0.9 comma negative 2.2.
- The curve then moves upward and to the right until it ends on the x axis at 2.

C.

- Two vertical dashed lines are shown at x equals negative 1 and x equals 1.
- The leftmost curve begins just above the x axis at x equals negative 2.
  - The curve moves steeply upward and to the right, becoming increasingly steep as it approaches the dashed vertical line at x equals negative 1.
  - The curve is nearly vertical as it ends, high above the x axis, just to the left of x equals negative 1.
- The second curve begins far below the x axis, just to the right of x equals negative 1.
  - The curve moves upward and to the right until it levels off and reaches a maximum on the y axis at negative 1.
  - It then moves downward and to the right until it ends just to the left of x equals 1 and far below the x axis.
- The third curve begins high above the x axis, just to the right of x equals 1.
  - The curve is nearly vertical initially.
  - It then moves steeply downward and to the right, levelling off as it approaches the x axis and ending just above the x axis at x equals 2.

D.

- The curve is nearly horizontal as it begins at a minimum at the point with coordinates negative 2 comma negative 2.

- It then moves steeply upward and to the right, crosses the x axis at negative 1, and ends at a maximum on the y axis at 2.
- The curve continues downward and to the right, crosses the x axis at 1, and is nearly horizontal as it ends at a minimum at the point with coordinates 2 comma negative 2.

---

594. For $t > 0$, a parametric curve in the *xy*–plane is defined by $x(t) = t^2 - 4t + 5$ and $y(t) = t^2 - 6t$. For what value of *t* does the curve have a vertical tangent?

A. 2

B. 3

C. 5

D. There is no such value of *t*.

---

595. Let *f* be a function with second derivative given by $f''(x) = (x-1)^2(x-2)(x-3)^3$. How many points of inflection does the graph of *f* have?

A. Two

B. Three

C. Five

D. Six

---

596. If at any time *t* the position vector of a particle moving in the *xy*–plane is $\langle \sin(\pi t), \cos(\pi t) \rangle$, what is the acceleration vector of the particle at time $t = \frac{1}{2}$?

A. $\langle -\pi^2, 0 \rangle$

B. $\langle 0, -\pi^2 \rangle$

C. $\langle -1, 0 \rangle$

D. $\langle 0, 1 \rangle$

---

597. What are all values of *p* for which the series $\sum_{n=1}^{\infty} \frac{1}{n^{4p-1}}$ converges?

A. $p \leq \frac{1}{2}$

B. $p \leq 1$

C. $p > \frac{1}{2}$

D. $p > 1$

---

598. If 10 people have heard the news at time $t = 0$, how many people have heard the news when the rate at which the news is spreading changes from increasing to decreasing?

The number of people at a business who have heard a piece of news at time *t* days is modeled by the function *P*. The rate at which the news is spreading is modeled by the logistic differential equation $\frac{dP}{dt} = 144P\left(1 - \frac{P}{4800}\right)$.

A. 72

B. 144

C. 2400

D. 4800

---

599. If $x^2 - xy - 3y^2 = 6$, then $\frac{dy}{dx} =$

A. $\frac{2x}{1+6y}$

B. $\frac{2x-y}{x+6y}$

C. $\frac{2x+y}{x+6y}$

D. $\frac{y-2x}{x+6y}$

---

600. What is the radius of convergence of the series $\sum_{n=0}^{\infty} \frac{3nx^n}{(n+2)^2}$?

A. $\frac{1}{3}$

B. 1

C. 3

D. Infinite

---

601. If the area of each region is 3, what is the value of $\int_0^4 f(x)dx + \int_0^7 f(x)dx$?

Regions *A* and *B* in the figure shown are bounded by the graph of the function *f* and the *x*–axis.

- The curve is labeled y equals f open parentheses x close parentheses and begins above the x axis, just to the left of the y axis, and moves downward and to the right.

- It passes through the origin then levels off as it reaches a local minimum slightly below the x axis at x equals 2.
- The curve then moves upward and to the right, crosses the x axis at 4 and levels off as it reaches a local maximum above the x axis at x equals 5.5.
- It then moves downward and to the right, crosses the x axis at 7, and continues downward and to the right until it ends at a point far below the x axis, far to the right of the y axis.
- The region bounded from above by the x axis and from below by the curve from x equals 0 to x equals 4 is labeled A.
- The region bounded from above by the curve from x equals 4 to x equals 7 and bounded from below by the x axis is labeled B.

A. -3

B. 0

C. 3

D. 9

---

602. If $y = \sin^{-1}(x-1)$, then $\frac{dy}{dx} =$

A. $\frac{1}{\sqrt{2-x}}$

B. $\frac{1}{\sqrt{1-x^2}}$

C. $\frac{1}{\sqrt{2x-x^2}}$

D. $\frac{1}{x^2-2x+2}$

---

603. Which of the following statements about the series $\sum_{n=1}^{\infty} \frac{n^2+n}{n^3+3n^2-1}$ is true?

A. The series converges because $\lim_{n\to\infty} \frac{n^2+n}{n^3+3n^2-1} = 0$.

B. The series converges by limit comparison to the series $\sum_{n=1}^{\infty} \frac{1}{n^3}$.

C. The series diverges by the ratio test.

D. The series diverges by limit comparison to the series $\sum_{n=1}^{\infty} \frac{1}{n}$.

---

604. Which of the following satisfies the differential equation $\frac{dy}{dx} = \cos\left(\frac{\pi}{2}x^2\right)$ with the initial condition $y(0) = 2$?

A. $y = 2 + \sin\left(\frac{\pi}{2}x^2\right)$

B. $y = 2 + \frac{1}{\pi x}\sin\left(\frac{\pi}{2}x^2\right)$

C. $y = \int_2^x \cos\left(\frac{\pi}{2}t^2\right)dt$

D. $y = 2 + \int_0^x \cos\left(\frac{\pi}{2}t^2\right)dt$

---

605. The second and third terms of the geometric series $\sum_{n=1}^{\infty} a_n$ are $a_2 = -\frac{3}{4}$ and $a_3 = \frac{1}{4}$, respectively. What is the sum of the series $\sum_{n=1}^{\infty} a_n$?

A. $-\frac{9}{16}$

B. $\frac{1}{16}$

C. $\frac{27}{16}$

D. $\frac{27}{8}$

---

606. Let $y = f(x)$ be the solution to the differential equation $\frac{dy}{dx} = 3y - 4x$ with initial condition $f(1) = 2$. What is the approximation for $f(2)$ if Euler's method is used, starting at $x = 1$ with a step size of 0.5?

A. 3

B. 4

C. 4.5

D. 5.5

---

607. Let *g* be the function given by $g(x) = \int_2^x f(t)dt$. At what value of *x* in the closed interval $[-3, 5]$ does *g* have an absolute minimum?

The graph of the function *f* on the closed interval $[-3, 5]$ is shown.

- The numbers negative 3 through 5, in increments of 1, are indicated on the horizontal axis.
- The numbers negative 2 through 3, in increments of 1, are indicated on the vertical axis.

- The graph consists of three line segments.
  - The first line segment begins at the point with coordinates negative 3 comma 2, and moves horizontally to the right until it ends at the point with coordinates negative 1 comma 2.
  - The second line segment begins where the first segment ends, and slants downward and to the right. It crosses the vertical axis at 1, crosses the horizontal axis at 1, and ends at the point with coordinates 3 comma negative 2.
  - The third line segment begins where the second segment ends, and slants upward and to the right. It crosses the horizontal axis at 4 and ends at the point with coordinates 5 comma 2.

A. -3

B. 1

C. 3

D. 4

---

608. $\int_2^3 \frac{4}{(x+3)(x-1)} dx =$

A. $\ln\left(\frac{3}{5}\right)$

B. $\ln\left(\frac{5}{3}\right)$

C. $\frac{1}{4}\ln\left(\frac{12}{5}\right)$

D. $4\ln\left(\frac{12}{5}\right)$

---

609. Let $y = f(x)$ be the particular solution to the differential equation $\frac{dy}{dx} = 3x^2y$ with initial condition $f(0) = 6$. Which of the following is an expression for $f(x)$?

A. e^{x^3}

B. $e^{x^3} + 5$

C. $e^{x^3} + 6$

D. $6e^{x^3}$

---

610. The line $y = 2$ is a horizontal asymptote to the graph of which of the following functions?

A. $y = \frac{x+2\sin x}{x-2}$

B. $y = \frac{2x^2+\sin x}{x^2}$

C. $y = \frac{2x^2+\sin x}{4-x^2}$

D. $y = \frac{2^x+2\sin x}{2^x}$

---

611. What is the perimeter of the region enclosed by the graphs of $y = 3$ and $y = x^2 - 2x$?

A. $4 + \int_{-1}^{3} \left(x^2 - 2x\right) dx$

B. $4 + \int_{-1}^{3} \sqrt{(2x-1)^2} dx$

C. $4 + \int_{-1}^{3} \sqrt{1 + (x^2 - 2x)^2} dx$

D. $4 + \int_{-1}^{3} \sqrt{1 + (2x-2)^2} dx$

---

612. If F is an antiderivative of a continuous function f, then $\int_{-1}^{2} f(3x+4) dx =$

A. $\frac{1}{3}F(10) - \frac{1}{3}F(1)$

B. $\frac{1}{3}F(2) - \frac{1}{3}F(-1)$

C. $F(10) - F(1)$

D. $F(2) - F(-1)$

---

613. $\lim\limits_{x \to \pi/4} \frac{\int_{\pi/4}^{x} \tan(t) dt}{x^2 - \frac{\pi^2}{16}}$ is

A. 0

B. $\frac{2}{\pi}$

C. $\frac{4}{\pi}$

D. nonexistent

---

614. $\int \frac{x^2+4x-1}{x+2} dx =$

A. $\frac{\frac{1}{3}x^3+2x^2-x}{\frac{1}{2}x^2+2x} + C$

B. $x + 2 - 5\ln|x+2| + C$

C. $\frac{1}{2}x^2 + 2x - 5\ln|x+2| + C$

D. $\frac{1}{2}x^2 + 2x + \frac{5}{2}(x+2)^{-2} + C$

---

615. $\int_1^\infty xe^{-x}dx$ is

A. 0

B. $\frac{1}{e}$

C. $\frac{2}{e}$

D. nonexistent

---

616. What is the slope of the line tangent to the polar curve $r = \theta^2$ at the point where $\theta = \pi$?

A. 2π

B. $\frac{\pi}{2}$

C. $\frac{2}{\pi}$

D. 0

---

617. What is the sum of the series shown?

$$1 - \frac{\left(\frac{\pi}{3}\right)^2}{2!} + \frac{\left(\frac{\pi}{3}\right)^4}{4!} - \frac{\left(\frac{\pi}{3}\right)^6}{6!} + \cdots + \frac{(-1)^n\left(\frac{\pi}{3}\right)^{2n}}{(2n)!} + \cdots$$

A. $\frac{1}{2}$

B. $\frac{\sqrt{3}}{2}$

C. $e^{-(\pi/3)^2}$

D. $\frac{1}{1+\frac{1}{2}\left(\frac{\pi}{3}\right)^2}$

---

618. Which of the following is the graph of a function *f* such that $\lim_{x\to 2^-} f(x)$ and $\lim_{x\to 2^+} f(x)$ both exist, but $\lim_{x\to 2} f(x)$ does not exist?

A.

- Two open circles and one closed circle are also in the graph.
- The first line segment begins below the x axis, to the left of the y axis, and slants upward and to the right. It crosses the origin and ends at an open circle above the x axis at x equals 2.

- A closed circle is indicated on the x axis at 2.
- The second line segment begins at an open circle below the x axis at x equals 2, and slants upward and to the right until it ends at a point above the x axis, far to the right of the y axis.

B.

- One open circle and one closed circle are also in the graph.
- The curve is nearly horizontal as it begins at a minimum point below the x axis, to the left of the y axis.
- It then moves steeply upward and to the right, crosses the origin, and ends at a maximum at an open circle above the x axis at x equals 2.
- The curve continues downward and to the right, crosses the x axis, and is nearly horizontal as it ends at a minimum point below the x axis, to the right of the y axis.
- A closed circle is indicated at x equals 2, between the x axis and the open circle at x equals 2.

C.

- The curve begins below the x axis, to the left of the y axis, and moves upward and to the right.
- It crosses the y axis below the origin and crosses the x axis at approximately 1.
- The curve becomes nearly vertical as it reaches x equals 2, above the x axis.
- The curve then moves nearly vertically downward initially, then downward and to the right.
- It crosses the x axis at approximately 3, and ends at a point below the x axis, to the right of the y axis.

D.

- A vertical dashed line is drawn at x equals 2.
- The first curve begins just above the x axis, to the left of the y axis.
    - The curve moves gradually upward and to the right at first.
    - It crosses the y axis just above the origin and becomes increasingly steep as it approaches the dashed vertical line at x equals 2.

- The curve is nearly vertical as it ends, high above the x axis, just to the left of x equals 2.

- The second curve begins high above the x axis, just to the right of x equals 2.
  - The curve is nearly vertical initially.
  - It then moves steeply downward and to the right, levelling off as it approaches the x axis.
  - The curve continues gradually downward and to the right until it ends at a point just above the x axis, far to the right of the y axis.

---

619. Which of the following gives the area of *R*?

Let *R* be the region bounded on the left by the graph of $x = f(y)$ and on the right by the graph of $x = g(y)$, as shaded in the figure.

- The graph shows a shaded region bounded on the left by the graph of x equals f open parentheses y close parentheses, and bounded on the right by the graph of x equals g open parentheses y close parentheses.
- The graph of each function is a curve.
- The graph of x equals f open parentheses y close parentheses begins slightly to the right of the y axis, below the x axis.
  - The curve moves upward and to the left until it becomes vertical as it reaches the point with coordinates 0 comma negative 1 on the y axis.
  - It then moves upward and to the right, crosses the x axis slightly to the right of the origin, and passes through the point with coordinates 2 comma 1.
- The graph of x equals g open parentheses y close parentheses begins slightly to the left of the y axis, below the x axis.
  - The curve moves upward and to the right.
  - It crosses the y axis and intersects the x equals f open parentheses y close parentheses curve at the point with coordinates 0 comma negative 1.
  - It continues upward and to the right, crosses the x axis to the right of the origin, and intersects the x equals f open parentheses y close parentheses curve again at the point with coordinates 2 comma 1.

A. $\int_{-1}^{1}(f(y) - g(y))dy$

B. $\int_{-1}^{1}(g(y) - f(y))dy$

C. $\int_0^2 (f(y) - g(y))dy$

D. $\int_0^2 (g(y) - f(y))dy$

---

620. Let *f* be a function that is continuous on the closed interval $[0, 2]$ with $f(0) = -2$, $f(1) = 5$, and $f(2) = 0$. Which of the following statements must be true?

A. The maximum value of *f* on the closed interval $[0, 2]$ is 5.

B. There is a number *c*, $0 < c < 2$, with $f(c) = -1$.

C. There is a number *c*, $0 < c < 2$, with $f(c) = 7$.

D. There is a number *c*, $0 < c < 2$, with $f'(c) = 1$.

---

621. Let *f* be the function with first derivative defined by $f'(x) = 3\cos\left(e^{2x}\right)$ on the interval $0 < x < 1.5$. What is the smallest value of *x* at which *f* has a relative minimum on $0 < x < 1.5$?

A. 0.226

B. 0.572

C. 0.775

D. 1.199

---

622. What is the total area of the region enclosed by the curve?

The figure shows the polar curve $r = 3\sin(5\theta)$ for $0 \leq \theta \leq \pi$.

- The curve consists of five elongated loops, each of which begins and ends at the origin.
  - One loop is located in quadrant 1.
  - One loop is located on the positive portion of the y axis.
  - One loop is located in quadrant 2.
  - One loop is located in quadrant 3.
  - One loop is located in quadrant 4.

A. 0.600

B. 1.200

C. 7.069

D. 14.137

---

623. Consider the series $\sum_{n=1}^{\infty} a_n$, where $a_n > 0$ for all *n*. Which of the following statements must be true?

A. If $\lim_{n\to\infty} a_n = 0$, then $\sum_{n=1}^{\infty} a_n$ converges.

B. If $\lim_{n\to\infty} a_n = 0$, then $\sum_{n=1}^{\infty} a_n$ diverges.

C. If $\lim_{n\to\infty} a_n = \frac{1}{2}$, then $\sum_{n=1}^{\infty} a_n$ converges.

D. If $\lim_{n\to\infty} a_n = \frac{1}{2}$, then $\sum_{n=1}^{\infty} a_n$ diverges.

---

624. Which of the following equations describes the situation?

The rate at which people register for a new social networking site is given by $g(t)$, where *t* is measured in minutes and $g(t)$ is measured in people per minute. The average rate of registration during the first 60 minutes is equal to the rate of registration at time $t = 30$ minutes.

A. $\int_0^{60} g(t)dt = g'(30)$

B. $\int_0^{60} g'(t)dt = g'(30)$

C. $\frac{1}{60}\int_0^{60} g(t)dt = g(30)$

D. $\frac{1}{60}\int_0^{60} g'(t)dt = g(30)$

---

625. At time $t \geq 0$, a particle moving in the *xy*–plane has a velocity vector given by $v(t) = \langle \sin(2t), e^t \rangle$. What is the total distance the particle travels over the interval $0 \leq t \leq 1$?

A. 1.536

B. 1.867

C. 2.185

D. 3.789

---

626. What is the volume of the solid formed when the region bounded by the graphs of $f(x) = \frac{1}{2}x^2 - x + 1$ and $g(x) = 5\sin\left(\frac{x}{3}\right)$ is revolved about the *x*–axis?

A. 5.006

B. 23.815

C. 26.738

D. 74.816

---

627. How many critical points does the function *f* have on the interval $0 < x < 14$?

The function *f* is continuous on the interval $0 < x < 14$. Shown in the figure is the graph of f', the derivative of *f*, on the interval $0 < x < 14$. The graph of f' is tangent to the *x*–axis at $x = 9$.

- The graph consists of two curves.
- Two open circles are included in the graph.
- The first curve begins on the y axis, below the origin.
    - The curve moves upward and to the right.
    - It reaches a local maximum just below the x axis, then moves downward and reaches a minimum.
    - The curve then moves upward and to the right, crosses the x axis, and ends above the x axis at an open circle at a point to the left of x equals 9.
- The second curve begins at a second open circle located directly above the first.
    - The curve moves downward and to the right until it reaches a minimum on the x axis at 9.
    - It then moves upward and reaches a maximum before moving briefly downward and to the right until it ends at x equals 14, and above the x axis.

A. One

B. Two

C. Three

D. Four

---

628. A particle moves along a straight line with velocity given by $v(t) = t^2 - 3t + e^{-t/2}$ and acceleration given by $a(t) = 2t - 3 - \frac{1}{2}e^{-t/2}$ for time $0 \le t \le 4$. On which open intervals is the speed of the particle increasing?

A. $(0, 0.318)$ and $(2.921, 4)$

B. $(1.612, 4)$ only

C. $(0, 0.318)$ and $(1.612, 2.921)$

D. $(0.318, 1.612)$ and $(2.921, 4)$

---

629. Using the third-degree Taylor polynomial for *f* about $x = 1$, what is the approximation for $f(0.7)$?

Let *f* be a function with derivatives of all orders for all values of *x*. Selected values of *f* and its first three derivatives are indicated in the table shown.

x	$f(x)$	$f'(x)$	$f''(x)$	$f'''(x)$
0	0.2	0	0	6
1	1	2.2	3	-2.3

A. 0.485

B. 0.496

C. 0.543

D. 0.672

---

630. If $\int_a^b 2f(x)dx + \int_c^b f(x)dx = 5$ and $\int_a^c f(x)dx = 1$, then $\int_a^b f(x)dx =$

A. 4

B. 2

C. 1

D. -1

---

631. At what rate is the volume of the cylinder changing, in cubic inches per minute, at that instant?

The volume of a cylinder with radius *r* inches is given by $V = \pi r^2\left(4r + \sin\left(0.2r^2\right)\right)$. At the instant when the radius is 2.7 inches, the radius is increasing at a rate of 0.75 inch per minute.

A. 202.575

B. 220.851

C. 270.101

D. 294.467

---

632. If $T_3(0.3)$ is used to approximate $f(0.3)$, what is the maximum possible error guaranteed by the Lagrange error bound?

Let *f* be a function that has derivatives of all orders for all real numbers, and let $T_3(x)$ be the third-degree Taylor polynomial for *f* about $x = 0$. The graph of $y = f^{(4)}(x)$, the fourth derivative of *f*, is shown for $0 \le x \le 0.3$.

- The graph is an upward-opening parabola.
- The parabola begins on the y axis at negative 2 and moves downward and to the right until it reaches a minimum at the point with coordinates 0.1 comma negative 3.
- It then moves upward and to the right, passes through the point with coordinates 0.2 comma negative 2 and ends at the point with coordinates 0.3 comma 1.

A. $\frac{1\cdot(0.3)^3}{3!}$

B. $\frac{3\cdot(0.3)^3}{3!}$

C. $\frac{1\cdot(0.3)^4}{4!}$

D. $\frac{3\cdot(0.3)^4}{4!}$

---

633. A tank contains 30 liters of water at time $t = 0$. For time $0 \le t \le 20$ minutes, water flows into the tank at a rate modeled by the function *f* defined by $f(t) = 2\ln(t+3) - 1.6$. For time $20 < t \le 45$ minutes, water flows into the tank at a rate modeled by the function *g* defined by $g(t) = \frac{7.5}{\ln t} + 2.2$. Both $f(t)$ and $g(t)$ are measured in liters per minute.

A. Find the rate of change of *g* at time $t = 23$ minutes. Show the setup for your calculations, and indicate units of measure.

B. How many liters of water are in the tank at time $t = 20$ minutes? Show the setup for your calculations.

C. Let *w* be the function defined by $w(t) = \begin{cases} f(t) & \text{for } 0 \le t \le 20 \\ g(t) & \text{for } 20 < t \le 45 \end{cases}$. Find $\lim_{t \to 20^+} w(t)$. Is *w* continuous at $t = 20$? Use the definition of continuity to justify your answer.

D. Consider the function *w* defined in part C. Find the average value of *w* from time $t = 15$ minutes to time $t = 30$ minutes. Show the setup for your calculations.

634.

A. Let *R* be the shaded region that is inside the curve $r = g(\theta)$ for $0 \le \theta \le 2\pi$. Find the area of region *R*. Show the setup for your calculations.

B. For each θ in the interval $0 \le \theta \le 2\pi$, the ray from the origin with angle θ intersects the curve $r = f(\theta)$ and the curve $r = g(\theta)$. For any value of θ, the distance between the curves is the difference between the *r*–values of the outer curve and the inner curve. Find the average distance between the two curves for $0 \le \theta \le 2\pi$. Show the setup for your calculations.

C. Find the value of θ in the interval $\frac{\pi}{4} \le \theta \le \frac{\pi}{2}$ that corresponds to the point on the curve $r = g(\theta)$ with greatest distance from the origin. Justify your answer.

D. There is a point *P* on the curve $r = f(\theta)$ where the slope of the line tangent to the curve at *P* is equal to $\sqrt{2} - 1$. At point *P*, the rate of change of the *x*–coordinate with respect to θ is $\frac{\sqrt{2}}{2}$. Find the rate of change of the *y*–coordinate with respect to θ at point *P*. Show the work that leads to your answer.

The graph shows the polar curves $r = f(\theta) = 1 + \sin\theta$ and $r = g(\theta) = 1 + \sin\theta + \cos^2(3\theta)$. The derivative of $g(\theta)$ with respect to θ is $g'(\theta) = \cos\theta - 3\sin(6\theta)$.

(Note: Your calculator should be in radian mode.)

- The curves are labeled r equals f open parentheses theta close parentheses and r equals g open parentheses theta close parentheses.

- The curve labeled r equals f open parentheses theta close parentheses begins at the origin.
    - The curve moves briefly downward and to the right until it reaches a minimum just below the x axis at approximately x equals one half.
    - It then moves upward and to the right, crosses the x axis at 1, and becomes increasingly steep until it is vertical at the point with approximate coordinates 1.2 comma 0.75.
    - It then moves upward and to the left until it levels off and reaches a maximum on the y axis at 2.
    - The curve then moves downward and to the left and becomes increasingly steep until it is vertical at the point with approximate coordinates negative 1.2 comma 0.75.
    - It then moves downward and to the right, crosses the x axis at negative 1, and levels off as it reaches a minimum just below the x axis at approximately x equals negative one half.
    - The curve then moves upward and to the right until it ends where it began at the origin.
- The curve labeled r equals g open parentheses theta close parentheses begins at the origin.
    - The curve moves steeply downward and to the right until it levels off and reaches a minimum at the point with approximate coordinates 0.6 comma negative 1.
    - It then moves upward and to the right and then left until it reaches a minimum point of the r equals f open parentheses theta close parentheses curve just below the x axis at approximately x equals one half.
    - It then moves to the right and curves slightly upward until it is nearly vertical as it crosses the x axis at 2.
    - The curve then moves upward and to the left until it is nearly vertical as it reaches the r equals f open parentheses theta close parentheses curve at the point with approximate coordinates 1.2 comma 0.75.
    - It then moves upward and to the right and then left until it reaches a maximum at the point with approximate coordinates 1.2 comma 2.5.
    - The curve then moves downward and to the left until it reaches the maximum of the r equals f open parentheses theta close parentheses curve on the y axis at 2.
    - It then moves upward and to the left until it reaches a maximum at the point with approximate coordinates negative 1.2 comma 2.5.

- The curve then moves downward and to the left and then right until it is nearly vertical as it reaches the r equals f open parentheses theta close parentheses curve at the point with approximate coordinates negative 1.2 comma 0.75.
- It then moves downward and to the left until it is nearly vertical as it crosses the x axis at negative 2.
- The curve then moves to the right until it reaches a minimum of the r equals f open parentheses theta close parentheses curve just below the x axis at approximately x equals negative one half.
- It then moves downward and to the left and then right until it reaches a minimum at the point with approximate coordinates negative 0.6 comma negative 1.
- It then moves upward and to the right until it ends where it began at the origin.

- The regions inside each curve are shaded.

635.

A. On what open intervals contained in $0 < x < 7$, if any, is the graph of *f* both increasing and concave down? Give a reason for your answer.

B. Find the average rate of change of *f* over the interval $0 \leq x \leq 4$. Can the Mean Value Theorem be applied to *f* on the interval $0 \leq x \leq 4$ to guarantee a value of *c*, for $0 < c < 4$, such that $f'(c)$ is equal to this average rate of change? Give a reason for your answer.

C. Use a left Riemann sum with the three subintervals indicated by the data in the table to approximate the value of $\int_0^7 f(x)dx$.

D. Let *g* be the function defined by $g(x) = x \cdot f(x)$. Is *g* increasing or decreasing at $x = 1$? Give a reason for your answer.

Let *f* be a continuous function defined on the interval $0 \leq x \leq 7$. The function *f* and its derivatives have the properties indicated in the table shown, where DNE indicates the *x*–values at which the derivatives of *f* do not exist.

x	$x = 0$	$0 < x < 2$	$x = 2$	$2 < x < 4$	$x = 4$	$4 < x < 7$	$x = 7$
$f(x)$	-1	Negative	-4	Negative	-2	Negative	0
$f'(x)$	DNE	Negative	DNE	Positive	0	Positive	DNE
$f''(x)$	DNE	Positive	DNE	Negative	0	Positive	DNE

---

636. For $0 \leq t \leq 120$, the temperature of a bowl of soup at time t is modeled by the function S that satisfies the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$, where $S(t)$ is measured in degrees Fahrenheit ($^\circ\mathrm{F}$) and t is measured in minutes. The temperature of the bowl of soup is always greater than $68^\circ\mathrm{F}$ and is $148^\circ\mathrm{F}$ at time $t = 0$.

A. Explain why the following could not be a slope field for the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$.

- The slope field consists of short line segments representing the slope at various points on the plane.
  - The line segments are horizontal at S equals 68.
  - As S increases, the line segments become increasingly steep upward and to the right.
  - As t increases, the behavior of the line segments is unchanged.

B. Use the line tangent to the graph of S at $t = 0$ to approximate the temperature of the soup at time $t = 5$ minutes.

C. According to the model, is the approximation found in part B an overestimate or an underestimate of the temperature of the soup at time $t = 5$ minutes? Give a reason for your answer.

D. Using separation of variables, find the particular solution $y = S(t)$ to the differential equation $\frac{dS}{dt} = -\frac{1}{20}(S - 68)$ with initial condition $S(0) = 148$.

637.

A. For each of $x''(2)$ and $x''(4)$, find the value or explain why it does not exist.

B. On what open intervals contained in $6 < t < 10$, if any, is the graph of $x(t)$ concave down? Give a reason for your answer.

C. A particle moving in the *xy*–plane has position $(x(t), y(t))$ with velocity vector $\langle x'(t), y'(t) \rangle$ for $0 \leq t \leq 10$, where $x'(t)$ is defined by the given graph and $y'(t) = te^{5t}$. The particle is at position $(5, 1)$ at time $t = 0$. Find the *y*–coordinate of the position of the particle at time $t = 10$. Show the work that leads to your answer.

D. For the particle defined in part C, find the speed of the particle at time $t = 2$. Show the work that leads to your answer.

Let x be a differentiable function of t for $0 \leq t \leq 10$. The graph of $x'(t)$, the derivative of $x(t)$, consists of two line segments and a semicircle, as shown in the figure.

- The graph consists of two line segments and a semicircle.
  - The first line segment begins at the origin and slants downward and to the right until it ends at the point with coordinates 4 comma negative 2.
  - The second line segment begins where the first segment ends, and slants upward and to the right. It crosses the horizontal axis at 5 and ends at the point with coordinates 6 comma 2.
  - The semicircle begins where the second segment ends and opens upward. It reaches a minimum on the horizontal axis at 8 and ends at the point with coordinates 10 comma 2.

---

638. The function f is defined by the power series $f(x) = \sum_{n=0}^{\infty} \frac{(-1)^n 2^n x^{2n}}{(2n+1)!} = 1 - \frac{2x^2}{3!} + \frac{4x^4}{5!} - \cdots + \frac{(-1)^n 2^n x^{2n}}{(2n+1)!} + \cdots$ for all real numbers x for which the series converges.

A. Use the ratio test to determine the interval of convergence of the power series for f.

B. Determine whether f has a relative minimum, a relative maximum, or neither at $x = 0$. Give a reason for your answer.

C. The first two nonzero terms of the power series for f are used to approximate $f\left(\frac{1}{2}\right)$. Use the alternating series error bound to determine an upper bound on the error of the approximation.

D. Write the first three nonzero terms and the general term for an infinite series that represents $\int_0^x f(t)dt$.